



Fig. 6: Photograph of the main PCB

PARTS LIST

Semiconductors:

T1	- 2N2222 NPN transistor
T2	- BC546 NPN transistor
D1, D2	- 1N5819 Schottky diode
D3	- 6A4 rectifier diode
LED1-LED40	- 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon, unless stated otherwise):

R1-R8	- 470-ohm
R8, R9	- 68-ohm, 1W
R11	- 0.33-ohm
R13	- 6.8-kilo-ohm
R14	- 1-kilo-ohm

Capacitors:

C1, C2	- 500F, 2.7V supercapacitor
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Miscellaneous:

SP1	- 17.5V, 10W solar panel
CON2	- 4-pin connector
CON3	- 3-pin connector
TCT1	- 50T primary and 50T secondary on toroidal core transformer
	- Brass lamp
	- PCBs
	- 12-pin terminal connector

Here, it is used to extract stored energy from the supercapacitors. Joule thief is a self-oscillating voltage booster circuit.

The Joule thief circuit here consists of NPN transistor T1 and ferrite-core transformer TCT1. The primary coil of the transformer is connected between capacitor voltage Vc and collector of transistor T1. Its secondary coil is connected to the base of T1 through current-limiting resistor R12.

Initially, with T1 switched on, current builds up in the primary coil of transformer and induces voltage

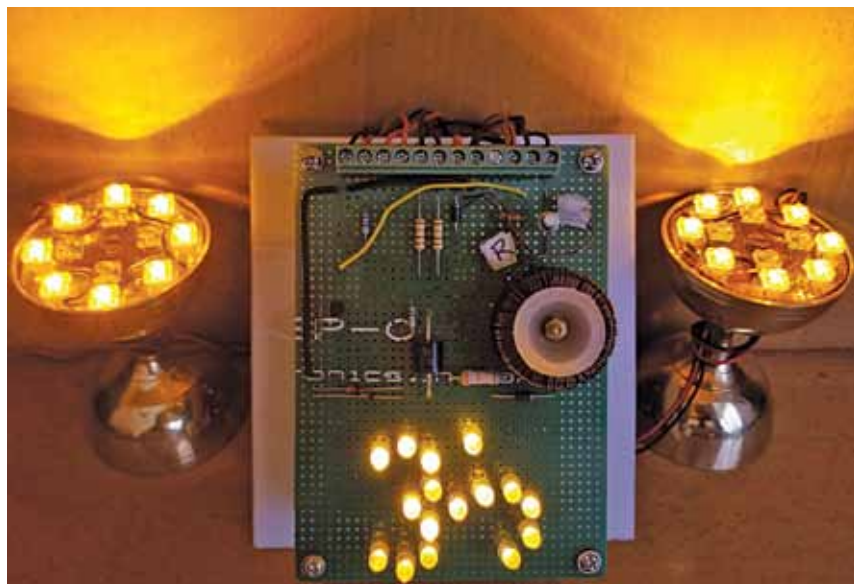


Fig. 7: Pooja lamp with daylamp LEDs on

in the secondary coil. The secondary coil drives the transistor to deep saturation. Once the current has built up in the primary, the current stops increasing. When there is no change of current occurring in the primary, there is no voltage induced in the secondary. This turns off T1. The stored energy in the primary produces a large voltage spike and the eight LEDs (LED33 through LED40) connected to the collector of T1 turn on. Once the stored energy in the primary is transferred to the LEDs, the transistor again turns on and the cycle repeats. Joule thief waveforms are shown in Fig. 4.

The eight LEDs that are powered using joule thief circuit are soldered on the octagonal PCB as shown in Fig. 3. LED33 through LED36 are soldered in series forming the inner circle LEDs on the left-side octagonal PCB and LED37 through LED40 are soldered in series on the right-side octagonal PCB.

Transistor T2 is used to sense the presence of voltage Vpv. Throughout the day, PV voltage is available and T2 is on. The collector of T2 is connected to base of T1. When T2 is on, the base of T1 is clamped to zero voltage. In the evening, when base voltage of T2 becomes $< 0.6V$, T2 turns off and joule thief circuit starts working and keeps the LEDs on through the night. Next morning, when PV voltage builds up, T2 turns on and joule thief circuit turns off. By then, the daylamp would have turned on. Thus, the proposed

pooja lamp circuit provides light without interruption.

The supercapacitors can be charged and discharged thousands of times, hence the circuit is likely to work for more than 15 years without any degradation in performance.

Construction and testing

The two supercapacitors C1 and C2 are mounted on a plastic base and connected in parallel as shown in Fig. 5. Fuse F1 is mounted on the PCB. Four standoffs are provided on the plastic base for mounting the main PCB.

Photograph of the main PCB is shown in Fig. 6. A 12-pin terminal connector is provided at the top edge of the PCB for all external connections. The capacitor charging circuit and the joule thief circuit are assembled on this PCB. LED17 through LED32 are also mounted on this PCB in the shape of 'Om.'

Fig. 7 shows pooja lamp with daylamp LEDs on. LED17 through LED32 are also on, which indicates that the capacitors are being charged. Fig.1 shows pooja lamp with joule thief LEDs turned on during night when PV voltage becomes zero. These lamps remain on throughout the night. **EFY**

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